

Autonomous-27 Rover: Docker Guide for Beginners

Why We Use Docker, How It Works, and Step-by-Step Setup for Every Teammate

1. Why Docker? (The Core Importance in Robotics)

Robotics development involves dozens of complex dependencies: specific ROS 2 distributions (Jazzy Jalisco), Gazebo Harmonic, OpenCV, PyTorch, Point Cloud Library (PCL), Patchwork++, RTAB-Map, and Nav2. Historically, setting this up directly on personal laptops caused the infamous **"It works on my machine"** problem, where differing Ubuntu versions, broken package conflicts, and driver incompatibilities consumed weeks of debugging.

Problem without Docker	How Docker Solves It for Autonomous-27
• Teammate A has Ubuntu 22.04 (Humble); Teammate B has 24.04 (Jazzy). • Missing Python/C++ libraries cause random build failures. • Pollutes personal laptops with gigabytes of conflicting system packages.	• 100% Identical Environment: Everyone runs the exact same ROS 2 Jazzy container. • Zero System Clutter: All dependencies stay neatly isolated inside the container. • Instant Onboarding: Clone the repo, run one script, and you are ready to code.

Key Advantages of Our Custom Docker Architecture:

- **Live Code Synchronization (Volume Mount):** Your project files on your laptop are mapped into /workspace. You edit code in your normal host IDE (VS Code, Cursor, CLion), and changes reflect inside Docker *instantaneously* without rebuilding images.
- **Native GUI & GPU Acceleration:** RViz2 and Gazebo Sim open natively as normal desktop windows using X11 socket sharing.
- **Host Networking:** ROS 2 DDS messages flow transparently between host, container, and physical rover sensors over Wi-Fi/Ethernet.

2. One-Time Host Setup (Run Once on Your Ubuntu Laptop)

Before using Docker, your host Linux system needs Docker and group permissions configured:

Step A: Install Docker & Docker Compose Plugin

```
sudo apt update && sudo apt install -y docker.io docker-compose-v2
```

Step B: Allow Running Docker Without 'sudo' (Crucial!)

By default, Docker requires root privileges. Add your user to the docker group so you never need to type sudo:

```
sudo usermod -aG docker $USER && newgrp docker
```

Note: If permission errors persist, simply log out of Ubuntu and log back in.

Step C (Optional): Enable NVIDIA GPU Hardware Acceleration

If your laptop has a dedicated NVIDIA graphics card, install the container toolkit for 60 FPS in Gazebo Sim & RViz:

```
sudo apt install -y nvidia-container-toolkit && sudo systemctl restart docker
```

3. How to Build & Run the Rover Container

Navigate to the repository's docker/ directory on your host terminal:

Step 1: Start the Docker Container (Terminal 1)

```
cd ~/Desktop/MESEKET/Autonomous-27/Autonomous_Ws/docker
./run_docker.sh
```

- *First run:* Docker downloads base images and installs ROS 2 Jazzy, Gazebo, OpenCV, and Patchwork++ (takes ~3-5 mins).
- *Subsequent runs:* Launches in **1 second** directly into root@hostname:/workspace#.

Step 2: Connecting More Terminals (Terminal 2, 3, 4...)

ROS 2 systems require multiple terminal tabs (one for simulation, one for perception, one for RViz). In every new terminal on your laptop, run:

```
cd ~/Desktop/MESEKET/Autonomous-27/Autonomous_Ws/docker
./enter_docker.sh
```

This connects you instantly to the active running container.

4. Built-in Terminal Shortcuts & Commands (Inside Container)

The container has pre-configured aliases to make daily workflow fast and enjoyable:

Alias	What it does	Standard ROS 2 Command
bld	Compile entire workspace with symlinks	colcon build --symlink-install
bldpkg <name>	Compile only one specific package	colcon build --packages-select <pkg>
sros	Source Jazzy & local install workspace	source /opt/ros/jazzy/setup.bash && source install/setup.bash
bldclean	Wipe cached build/install/log files	rm -rf build/ install/ log/
sim	Launch Mars Yard World & Rover	ros2 launch my_robot_description gazebo.launch.py
rviz	Launch RViz2 Visualization	ros2 launch my_robot_description rviz_only.launch.py
teleop	Drive Rover with keyboard keys	ros2 run teleop_twist_keyboard teleop_twist_keyboard
help	Print cheatsheet of all aliases	cat ~/.bash_aliases

5. Step-by-Step Daily Workflow for Team Members

- 1. Compile Workspace:** Run bld, then sros.
- 2. Terminal 1 (Simulation):** ros2 launch my_robot_description gazebo.launch.py world:=world1.world
- 3. Terminal 2 (Perception):** ros2 launch terrain_geometry terrain.launch.py
- 4. Terminal 3 (RViz):** ros2 launch terrain_geometry rviz.launch.py
- 5. Terminal 4 (Teleop):** teleop (use arrow keys / u-i-o-j-k-l to steer the rover towards rocks).

■ **3 Golden Rules for Beginners:**

- 1. Edit on Host, Run in Docker:** Open VS Code on your personal laptop normally. Never edit files via nano inside Docker unless quick debugging.
- 2. Remember to Source:** Every time you open a terminal or rebuild code, run sros so ROS knows about newly compiled nodes.
- 3. Exiting Docker:** Type exit to leave a terminal. The main container stays running in the background until all terminals close.